

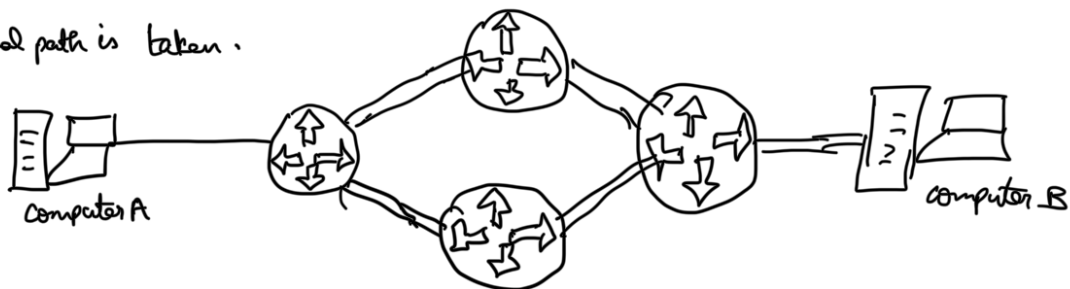
What is a Router?

* It's a router job to connect networks and pass data between them. It does this by using routing (hence the name router).

* Routing is label given to process of data travelling across networks. Routing involves creating a path between networks so that this data can be successfully delivered.

* Router operates at layer 3 of OSI model.

* Routing is useful when devices are connected by many paths, such as example below, where most optimal path is taken.



* Routers are dedicated devices and do not perform same functions as switches.

* Computer's A network is connected to network of Computer B by 2 routers in middle.

What path should be taken factors include:

- what path is shortest?
- what path is most reliable?
- which path is faster medium (eg:- copper or fibre)?

Switch?

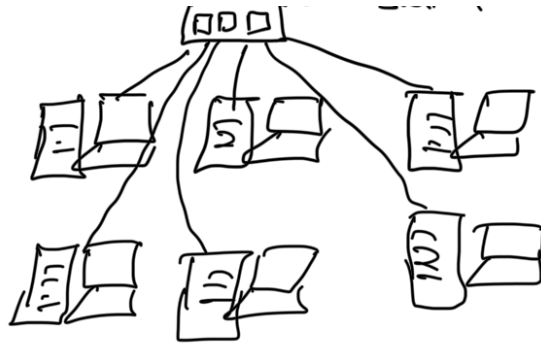
Switch is dedicated networking device responsible for providing means of connecting to multiple devices.

* Switch can facilitate many devices (3 to 63) using Ethernet cables.

* Switch can operate both layer 2 and layer 3 of OSI model. Layer 2 switches cannot operate at layer 3.

* Switch will forward frames onto connected devices using their MAC address.



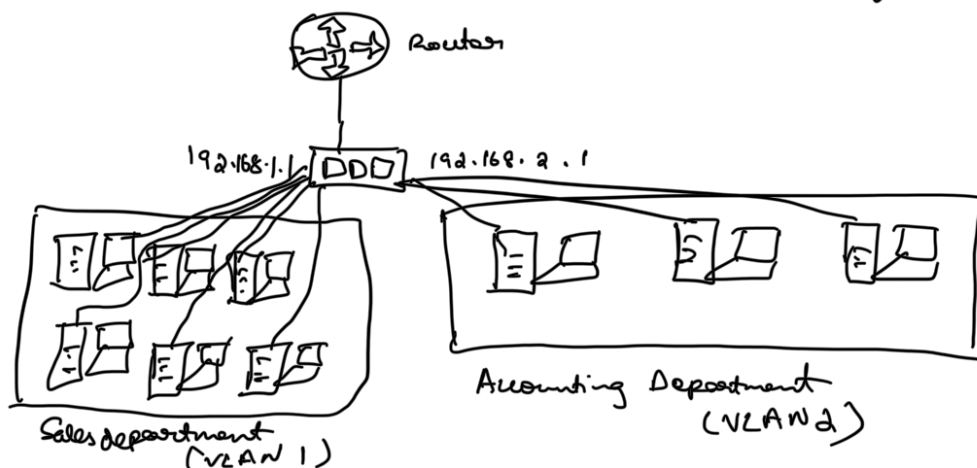


Now let's move to layer 3 switches, These switches are more sophisticated than layer 2, as they can perform some of responsibilities of router. Namely these switches will send frames to devices (as layer 2 does) and route packets to other devices using IP protocol.

Let's take a look at layer 3 diagram switch. We can see there are 2 IP addresses.

1) 192.168.1.1 2) 192.168.2.1

A technology called VLAN (virtual local area network) allows specific devices within network to be virtually split up. This split means they can all benefit from things such as Internet connection but are treated separately.



In context of diagram Sales Department and Accounting Department will be able to access the Internet, but not able to communicate with each other (although they are connected to same Switch).

Q) what is the verb for action that a router does?

A) Routing

Q) What are 2 different layers of switches? Separate these by commas i.e layer 2 and layer 3.

A) layer 2 and layer 3